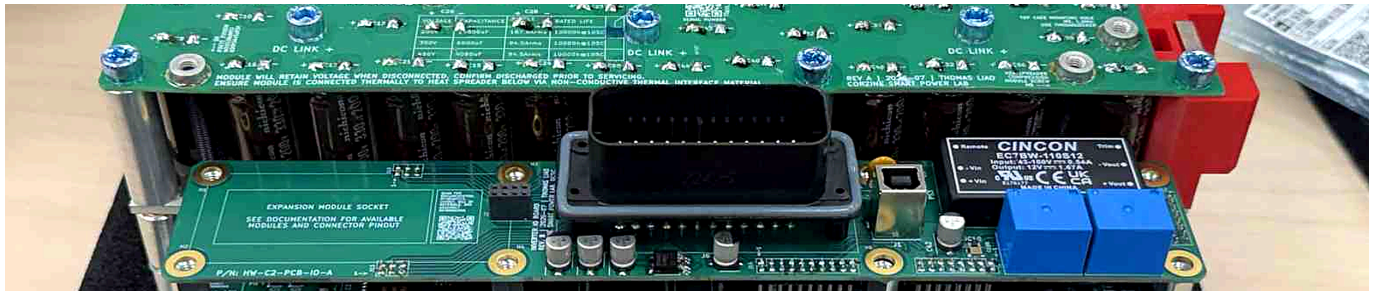


The open, safety-engineered power conversion platform

One reusable dual-MCU control module. Four power-stage chassis from ~50 kW to megawatt class. Motor drive to bidirectional DC-DC — IGBT today, SiC-ready. Every schematic, line of firmware, safety analysis, and test trace published.



The Size 2 control and power stack — built, bench-tested, and under validation.

400 V class

150 / 200 / 400 V by DC-link selection

465 A

Continuous · 600 A peak (60 s)

10

Published test reports, raw telemetry linked

PMSM + induction

Motor self-commissioning, instrument-verified

■ Built and bench-tested

Size 2 (~100 kW class) has run a **180 V bring-up**, a **one-hour endurance run**, a **20-minute reversal stress test**, and completed instrument-verified **motor self-commissioning**. Next: a Size 2 chassis for our 2018 Zero DSR electric motorcycle — integration in progress.

■ Safety engineered

Dual-MCU architecture with **six redundant safe-state pathways**. Hazard analysis (HARA) and FMEA with **ISO 26262** and **IEC 61800-5-2** mapping — every hazard traces to a safety requirement and a test.

■ Open everything

Schematics, firmware, the **RTE Studio** model-based workflow (no MATLAB; **alternative control algorithms benchmarked against FOC**), safety analyses, and **interactive telemetry logs** — all public at docs.openvvvf.org.

PLATFORM FAMILY

One control module — four power stages

CONTROL MODULE Dual-MCU · FOC motor control + independent safety coprocessor · USB-B / dual CAN FD / expansion · RTE Studio + Telemetry Viewer



Size 1

~50 kW class inverter for light EVs and lab benches

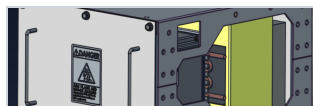
■ **EARLY DESIGN**



Size 2 Inverter

~100 kW · 400 V / 465 A — built and bench-tested

■ **IN VALIDATION**



Size 2 DC-DC

Bidirectional DC-DC for charging, storage, and microgrids

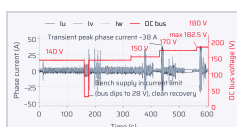
■ **IN PROGRESS**



Size 3

Megawatt class · 900 V / 1,400 A · liquid-cooled

■ **IN DESIGN**



Live telemetry, not promises

Test reports publish their interactive data logs — viewable in the browser-based Telemetry Viewer. Scan to open the 180 V bring-up: bus stepped 140 to 180 V, ~38 A peak, clean recovery from a bench-supply current-limit event.



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OpenVVF

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